

purdue-l3-site-operations brief

Purdue Site Operations (L3) — Cross-Vendor Threat & Detection Brief

Scope: This brief covers Purdue Level 3 (Site Operations) — the "domain controller" tier of OT. Roles in scope: SCADA servers, plant historians, engineering workstations (EWS), asset/configuration management servers, power-monitoring SCADA, substation Industrial PCs, console aggregators, and Windows-side power-management agents. Source material is aggregated from the [Eaton brief](#) (Group F Windows-side), [APC brief](#) (Group D PowerChute / EcoStruxure IT), [Vertiv brief](#) (Group A ACS 8000 + Group F DSView/Trellis), [SEL brief](#) (Group A SEL Industrial PCs + Group B acSEerator / BaRT / Compass / VPS), [Siemens brief](#) (TIA Portal V21 + WinCC), [Honeywell brief](#) (Experion PKS + Forge), [Schneider brief](#) (Control Expert, Citect, Power Operation, Vijeo), and [Rockwell brief](#) (Groups D + E: FactoryTalk + Studio 5000). Honeywell L3 content is research-only — firmware extraction is pending and is explicitly marked as such throughout.

Architecture grouping (organized by ROLE within this layer, not by vendor)

Class (role)	Products (cross-vendor)	Stack	Catalog depth
A. SCADA / HMI Server	Siemens WinCC + WinCC Unified; Rockwell FactoryTalk View SE / View ME; Schneider Citect SCADA / Plant SCADA; Schneider EcoStruxure Power Operation / Power Monitoring Expert; Schneider Vijeo Designer (Operator Terminal Expert); Honeywell Experion PKS Server	Windows Server + .NET + MSSQL + Tomcat / IIS; RNA / CTAPI / OPC UA backends	see siemens brief Group B , rockwell brief Group D , schneider brief Groups B/D/E , honeywell brief Group A
B. Engineering Workstation (EWS)	Siemens TIA Portal V21 (STEP7 + Openness + ALM); Rockwell Studio 5000 + RSLinx + FactoryTalk Linx; Schneider EcoStruxure Control Expert; SEL acSEerator QuickSet + BaRT + Compass + Virtual Port Service 5828; SEL-3355-2 / 3355 / 3300 Industrial PCs	Windows 10 IoT / Win Server, .NET, MSI/MSP, vendor IDE + project-file handlers	see siemens brief Group A , rockwell brief Group E , schneider brief Group B , sel brief Groups A+B

Class (role)	Products (cross-vendor)	Stack	Catalog depth
C. Asset / Config Management Server	Rockwell FactoryTalk AssetCentre; Rockwell FactoryTalk Historian SE; Rockwell FactoryTalk Linx Gateway; Vertiv Avocent DSView; Vertiv Trellis Platform; Vertiv Environet Alert; Vertiv Power Insight	Windows Server + MSSQL / Postgres + Java/Tomcat, holds crown-jewel PLC project archives + credentials	see rockwell brief Group D , vertiv brief Group F
D. Windows-side Power-Mgmt Agent	Eaton UPS Companion, Eaton IPM/IPP, Eaton setUPS, Eaton MIBs/PX-UPS/RNDIS drivers; APC PowerChute Network Shutdown, PowerChute Business / Personal; Schneider EcoStruxure IT Gateway / Expert	Windows .NET / Java services that broker between OS and UPS NMC	see eaton brief Group F , apc brief Group D
E. Console / Serial Aggregator	Vertiv Avocent ACS 8000 (4/8/16/32/48-port); SEL Virtual Port Service 5828 (serial-over-TCP)	ARM Linux + busybox + portmgr / cycladesd; Windows TCP-serial bridge	see vertiv brief Group A , sel brief Group B

Group A — SCADA / HMI Server

Direct attack surface (cross-vendor):

WinCC CCEServer/CCAgent • WinCC Unified Web Runtime (HTTPS/443) • OPC UA (TCP/4840) • MSSQL backend (Stuxnet's path) • Rockwell RNA + FactoryTalk Linx (TCP/3060 + dynamic) • Citect CTAPI (TCP/5482) + IOserver • Cicode runtime • Experion Control Data Access (UDP/55555) • WebNavigator IIS app • Schneider OFS OPC server • Vijeo Designer runtime download (TCP/6000)

Confirmed CVEs across vendors:

CVE	CVSS	Vendor	Product	Vector
Siemens ProductCERT WinCC web-client auth class	9.x	Siemens	WinCC	Improper auth in WinCC web client (V7.x family)
Siemens ProductCERT WinCC Unified Web Runtime class	8.x	Siemens	WinCC Unified	Path traversal in Unified Web Runtime
Siemens ProductCERT WinCC installer DLL-hijack class	7.8	Siemens	WinCC	DLL hijack on install

CVE	CVSS	Vendor	Product	Vector
CVE-2024-21914 (ICSA-24-018-02)	6.5	Rockwell	FactoryTalk View SE	DoS via crafted message
Rockwell PSIRT — View SE project-import RCE family	high	Rockwell	FactoryTalk View SE	RCE via crafted HMI project import
ICSA-23-193-01	varies	Rockwell	FactoryTalk View ME	Multiple unauth issues
CVE-2008-2639	10.0	Schneider	CitectSCADA	ODBC server stack overflow (Core Security)
CVE-2020-7548	7.8	Schneider	Citect Anywhere	Auth bypass
Schneider PSIRT — Plant SCADA path traversal family	n/a	Schneider	Plant SCADA (EcoStruxure)	Authenticated path traversal
CVE-2021-22722	7.5	Schneider	EcoStruxure Power Monitoring Expert	Improper privilege management
Schneider PSIRT — PME report-generator RCE family	n/a	Schneider	Power Monitoring Expert	RCE via report generator
CVE-2021-38397	10.0	Honeywell	Experion PKS / LX / PlantCruise	Unrestricted file upload → unauth RCE (research only)
CVE-2021-38395	9.1	Honeywell	Experion PKS / LX / PlantCruise	Argument injection in Honeywell-signed binary (research only)
ICSA-21-294-02	Critical	Honeywell	Experion PKS Server	Upload / argument injection / DLL-hijack cluster (research only)
ICSA-23-061-02	9.x	Honeywell	Experion PKS / LX / PlantCruise	ICEFALL "Crystallized Insecurity" cluster (research only)

Top attack vector (MITRE ATT&CK ICS): [T0832 Manipulation of View](#) + [T0859 Valid Accounts](#) on the SCADA SQL backend (the Stuxnet WinCC path) — tamper tag values shown to the operator while logic runs free.

Group B — Engineering Workstation (EWS)

Direct attack surface (cross-vendor):

TIA Portal Openness API (.NET, PowerShell-callable) · Automation License Manager (TCP/4410) · project-file handlers (.ap21/.zap21, .stu/.xef/.zef, .ACD, .aclx, .RDB) · Studio 5000 + RSLinx · FactoryTalk Services Platform · Control Expert + Vijeo Designer · acSEerator QuickSet · SEL-BaRT (Fast Message + ASCII serial-over-TCP firmware push) · SEL Compass updater · Virtual Port Service 5828 TCP listener (serial bridge) · SMB · RDP · WinRM

Confirmed CVEs across vendors:

CVE	CVSS	Vendor	Product	Vector
CVE-2024-49775	7.8	Siemens	TIA Portal (UMC)	Local privilege escalation via User Management Component
Siemens ProductCERT TIA Portal Openness class	7.x	Siemens	TIA Portal Openness	Project-file deserialization → code exec
Siemens ProductCERT Automation License Manager class	7.x	Siemens	Automation License Manager	LPE on ALM service
CVE-2021-22681	10.0	Rockwell	Logix / Studio 5000	Hardcoded crypto key → remote auth bypass to controller
CVE-2022-1161	7.7	Rockwell	Studio 5000 / CompactLogix 5380, ControlLogix 5580	Stealth logic injection (view ≠ controller bytecode)
CVE-2024-45824	9.8	Rockwell	FactoryTalk View ME / Studio 5000	Remote code injection via crafted project
CVE-2023-46290	7.5	Rockwell	FactoryTalk Services Platform	Improper auth, unauth user enumeration
CVE-2024-21915	9.8	Rockwell	FactoryTalk Service Platform	Privilege escalation, affects Logix sessions
Rockwell PSIRT — Studio 5000 DLL hijack family	high	Rockwell	Studio 5000 Logix Designer	DLL hijack on launch
Claroty "Evil PLC" — Studio 5000	n/a	Rockwell	Studio 5000	Weaponized PLC project file → RCE on EWS
CVE-2020-7559	7.8	Schneider	EcoStruxure Control Expert	DLL hijack at engineering workstation startup
CVE-2021-22779	8.1	Schneider	Control Expert ↔ PLC	UMAS session hijack from EWS (ModiPwn)

CVE	CVSS	Vendor	Product	Vector
Schneider PSIRT 2023 Control Expert project-file RCE	n/a	Schneider	EcoStruxure Control Expert	RCE via crafted .stu project (Armis-class)
Claroty "Evil PLC" — Control Expert	n/a	Schneider	Control Expert	Weaponized PLC pushes payload to EWS on upload
ICSA-23-131-08	n/a	SEL	acSElerator QuickSet (SEL-5030), RTAC (SEL-5033), ICS Studio (SEL-5045)	CWE-276 insecure perms + CWE-22 path traversal + CWE-269 improper privilege mgmt
SEL Security Advisories index	n/a	SEL	acSElerator QuickSet	XXE on .ac1x import → local file disclosure / SSRF
SEL Security Advisories index	n/a	SEL	SEL-5037 Grid Configurator	Cleartext / weakly-protected credential storage
SEL Security Advisories index	n/a	SEL	acSElerator Architect (SEL-5032)	Authorization / project-import handling weaknesses
ICSA-23-194-02	n/a	SEL	RTAC + co-installed mgmt binaries	Multiple weaknesses in RTAC firmware and mgmt interfaces

Top attack vector (MITRE ATT&CK ICS): [T0843 Program Download](#) — the EWS is the single highest-value pivot in any OT network because it holds project files and the credentials/certs to push logic. Chains with [T0873 Project File Infection](#) and [T0853 Scripting](#) (Openness API / Cicode payloads in tampered project archives — Stuxnet shape, modernised). Forescout's OT:ICEFALL and Claroty's "Evil PLC" both centred this surface.

Group C — Asset / Config Management Server

Direct attack surface (cross-vendor):

FactoryTalk Directory + AssetCentre (project archives + encrypted PLC credential vault) •
 FactoryTalk Historian SE • FactoryTalk Linx Gateway (TCP/3060) • DSVIEW mgmt-plane (Java) •
 Trellis Platform (Apache Tomcat + Postgres) • Environet Alert • Power Insight •
 MSSQL backends • IIS / Tomcat admin endpoints • OPC HDA + A&E

Confirmed CVEs across vendors:

CVE	CVSS	Vendor	Product	Vector
Rockwell PSIRT — AssetCentre deserialization	high	Rockwell	FactoryTalk AssetCentre	RCE via deserialization of project archive

CVE	CVSS	Vendor	Product	Vector
RCE family				
CVE-2024-21915	9.8	Rockwell	FactoryTalk Service Platform	Privilege escalation across FT user store
Vertiv PSIRT — Avocent DSView advisories	n/a	Vertiv	Avocent DSView	Pre-auth deserialization / RCE per vendor advisory
CVE-2021-44228	10.0	Vertiv	Trellis 5.x (Tomcat)	Apache Log4j JNDI RCE (Log4Shell) on unpatched Trellis
Synopsys CyRC — Avocent / HPE KVM advisory	n/a	Vertiv	Avocent KVM family (shared firmware)	Hardcoded creds + buffer overflow (unauth → root)

Top attack vector (MITRE ATT&CK ICS): [T0866 Exploitation of Remote Services](#) → [T0852 Screen Capture / Theft of Operational Information](#). AssetCentre / DSView / Trellis hold credentials and trust to every device they manage — single compromise yields every PLC password and KVM session in the plant.

Group D — Windows-side Power-Management Agent

Direct attack surface (cross-vendor):

Eaton setUPS.exe / IPM / UPS Companion installer + LocalSystem services •
APC PowerChute Java mgmt agent + IIS local web mgmt • EcoStruxure IT Gateway (Java) •
HTTPS / SNMP outbound to NMC fleet • DLL search-order surface in installer paths

Confirmed CVEs across vendors:

CVE	CVSS	Vendor	Product	Vector
CVE-2025-59887	8.6	Eaton	UPS Companion <3.0 (all versions)	DLL hijack RCE in installer
CVE-2025-59888	6.7	Eaton	UPS Companion <3.0	Unquoted service path → SYSTEM
CVE-2020-6650	7.x	Eaton	UPS Companion <1.06	Plaintext HTTP update + eval() of response
IPM pre-1.69 (ICSA-21-110-06)	9.8	Eaton	Intelligent Power Manager	Unauth eval injection RCE + unauth file upload + SQLi

CVE	CVSS	Vendor	Product	Vector
CVE-2021-22812	9.8	APC / Schneider	PowerChute Business Edition	Unauth RCE via deserialization in management agent
CVE-2021-22813	7.5	APC / Schneider	PowerChute Business Edition	Auth bypass on local web mgmt
Schneider PSIRT — PowerChute Serial Shutdown LPE class	7.x	APC / Schneider	PowerChute Serial Shutdown	Local privesc via insecure dir / service-permissions
Schneider PSIRT — EcoStruxure IT Gateway mgmt-plane class	7.x	APC / Schneider	EcoStruxure IT Gateway	Auth / management-plane vuln class

Top attack vector (MITRE ATT&CK ICS): [T1574.001 DLL Search Order Hijacking](#) (Enterprise) chained into [T0857 System Firmware](#) — admin host runs a tampered installer → SYSTEM → push tampered firmware via the legitimate update channel to every UPS NMC the agent trusts.

Group E — Console / Serial Aggregator

Direct attack surface (cross-vendor):

Vertiv ACS 8000: sshd • lighttpd HTTPS • portslave / cycladesd serial multiplexer • snmpd v1/v2c/v3 • strongSwan ipsec • root/admin default accounts.
SEL VPS 5828: Windows TCP listener bridging to local COM ports • default bind / no auth class.

Confirmed CVEs across vendors:

CVE	CVSS	Vendor	Product	Vector
ICSA-23-285-06	n/a	Vertiv	Avocent ACS 8000 firmware ≤3.3.x	OS command injection + auth bypass + hardcoded root cred
Vertiv PSIRT — ACS 8000 advisories	n/a	Vertiv	Avocent ACS 8000	Hardcoded credential / insecure default
Nozomi Networks Labs — ACS 8000 research	n/a	Vertiv	Avocent ACS 8000	Chained unauth → root via web UI + serial daemon
ICSA-23-131-08	n/a	SEL	Virtual Port Service 5828 (co-installed mgmt class)	DLL search-order / writable install-path → LPE to SYSTEM (acSElerator family covered by same advisory)

Top attack vector (MITRE ATT&CK ICS): [T0822 External Remote Services](#) + [T0886 Remote Services](#) — every serial port the aggregator brokers becomes a console-level pivot into a downstream device, bypassing that device's own network ACLs. Honour ICSA-23-285-06's "blast radius = every cabled device" framing.

Logging matrix (highest priority for this layer)

#	Source	Event	What it catches	MITRE ATT&CK ICS
1	Sysmon on EWS hosts (TIA Portal / Studio 5000 / Control Expert / acSElerator / BaRT)	EventID 1 ProcessCreate where ParentImage = <code>Siemens.Automation.*</code> / <code>RSLogix5000.exe</code> / <code>Studio5000.exe</code> / <code>RSLink.exe</code> / <code>ControlExpert.exe</code> / <code>AcSEleratorQuickSet.exe</code> / <code>BaRT.exe</code> / <code>Compass.exe</code> AND child = <code>cmd.exe</code> / <code>powershell.exe</code> / <code>rundll32.exe</code>	Evil-PLC / Openness-API abuse / scripted PLC push	T0853 Scripting → T0843 Program Download
2	Sysmon on EWS / SCADA / asset-mgmt hosts	EventID 7 ImageLoad — non-vendor DLLs loaded by <code>CCEServer.exe</code> / <code>Siemens.Automation*.exe</code> / <code>FTAC.exe</code> / <code>ViewSE*.exe</code> / <code>Studio5000.exe</code> / <code>ControlExpert.exe</code> / <code>AcSElerator*.exe</code> / <code>setUPS*.exe</code> / <code>PowerChute*.exe</code> from user-writable paths	DLL search-order hijack class (WinCC installer, Studio 5000, Eaton UPS Companion CVE-2025-59887, acSElerator ICSA-23-131-08, Schneider CVE-2020-7559)	T1574.001 DLL Search-Order Hijack
3	Sysmon EventID 11 (FileCreate) on EWS	<code>.stu</code> / <code>.xef</code> / <code>.zef</code> / <code>.ACD</code> / <code>.aclx</code> / <code>.RDB</code> / <code>.crd</code> / <code>.ctz</code> / <code>nmc*.bin</code> / <code>apc_hw*.bin</code> created outside project directory or by non-engineer accounts	Project-file infection / staged tampered firmware (Evil-PLC, Cicode payload, NMC firmware staging)	T0873 Project File Infection + T0857 System Firmware
4	FactoryTalk Diagnostics + AssetCentre audit	FT Directory <code>LogOn</code> / <code>LogOff</code> correlated with AssetCentre <code>CheckOut</code> of PLC project outside change-ticket window	Attacker pulling project to weaponize / credential mother-lode access	T0866 Exploitation of Remote Services
5	WinCC / Experion / Citect audit logs	User logon failed × N then success; Project loaded / PointParameterChange / Configuration changed outside maintenance window	Brute force on SCADA + unauthorised project load + operator-action abuse	T0859 Valid Accounts + T0832 Manipulation of View

#	Source	Event	What it catches	MITRE ATT&CK ICS
6	Network IDS (Suricata / Zeek)	S7comm-Plus, UMAS function 0x5A, CIP service codes 0x4B/0x4C/0x4D (program download) + 0x6B (mode change), DNP3 FC 5/6, IEC 61850 MMS <code>write</code> , GOOSE from non-relay MACs — all sourced from any host outside the EWS VLAN	Cross-vendor unauthorised PLC programming / mode flip / breaker tamper	T0843 Program Download + T0855 Unauthorized Command Message
7	Network IDS	Connections to Virtual Port Service 5828 TCP listener or ACS 8000 portmgr from non-loopback / non-engineer IPs; <code>login.success</code> for <code>root</code> on ACS 8000 from any IP	Lateral relay/console access via serial-tunnel pivot	T0822 External Remote Services + T0886 Remote Services
8	Sysmon on power-mgmt + KVM-mgmt hosts (PowerChute / DSView / Trellis / Tomcat)	EventID 1 where ParentImage = <code>tomcat*.exe</code> / <code>PowerChute*.exe</code> / <code>setUPS*.exe</code> / <code>IPM*.exe</code> AND child not in vendor allow-list	DSView pre-auth deserialization, Log4Shell on Trellis, PowerChute CVE-2021-22812 deserialization, Eaton UPS Companion installer RCE post-ex	T1059 Command and Scripting Interpreter (Enterprise) + T0857 System Firmware

Secondary:

- Firewall egress: any outbound from L3 SCADA / EWS / asset-mgmt VLANs to non-RFC1918 destinations besides documented vendor cloud endpoints (Forge Azure tenants, Compass `updates.selinc.com`, Rockwell PSIRT mirror). Volt Typhoon's 2024 US-infrastructure campaign (CISA AA24-038A) used exactly this kind of callback path off the EWS.
- Vendor-native audit feeds (turn on + forward via syslog): WinCC audit log; Experion Operator Action Journal + System Event Journal + Control Builder change tracking; FactoryTalk Diagnostics (`Audit`, `Security`, `Configuration` categories); SEL relay SER buffer / Audit log via Fast Message `HIS` / `MET`; ACS 8000 Data Buffering + Auditing.
- Subscribe to ProductCERT-feeds with monthly review jobs: Siemens ProductCERT (2nd Tuesday), Rockwell Trust Center RSS, Schneider PSIRT bulletins, Honeywell PSIRT, SEL Security Advisories, Eaton PSIRT, APC / Schneider, Vertiv PSIRT.
- Active Directory: monitor service accounts used by FactoryTalk, Citect, WinCC, Power Operation, ACS 8000 RADIUS clients for interactive logons — they should be service-only.

Cross-layer pivots

1. **L3 EWS → L1 PLC via authenticated CIP / S7Comm-Plus / UMAS write (Stuxnet model).**
Compromise the engineering workstation (Studio 5000, TIA Portal Openness, Control Expert, acSELeator BaRT), then use the vendor's own programming protocol from a legitimate session to push tampered logic. Chains DLL-hijack + project-file infection on the EWS into [T0843 Program Download](#) at L1. CVE-2021-22681 (Rockwell hardcoded crypto key), CVE-2022-38465 (Siemens S7-1500 global private key), and CVE-2021-22779 (Schneider ModiPwn) all reduce this to a single-step pivot once the EWS is owned.
2. **L3 historian → L4 corporate (the IT/OT data path attackers ride backward).** FactoryTalk Historian SE, WinCC archives, and Experion Forge OPC UA bridges all push plant data northbound through documented IT/OT seams. The same TLS / MQTT / Tomcat endpoints that publish historian data are the soft re-entry point: Log4Shell on Trellis ([CVE-2021-44228](#)), Forge edge-agent OPC UA client misconfiguration, and FactoryTalk Service Platform privilege escalation ([CVE-2024-21915](#)) all enable corporate-to-historian pivots that then reach the L2 HMI / L1 controller plane. Maps to [T0884 Connection Proxy](#) + [T0866 Exploitation of Remote Services](#).
3. **L3 AssetCentre → L1 mass-PLC config push (legitimate workflow, weaponized for tampering).** FactoryTalk AssetCentre is the disaster-recovery vault — it holds every PLC project archive + credentials in the plant. The Rockwell PSIRT AssetCentre deserialization family + [CVE-2024-21915](#) yield one-shot access to the entire fleet's program-download credential bundle, then attacker pushes tampered logic via the *normal* DR-restore workflow. Same model applies to SEL Compass-managed BaRT trees (silent relay-firmware tamper) and Schneider Secure Connect Advisor (cloud-pair token → on-prem PLC). [T0857 System Firmware](#) at fleet scale.
4. **L3 SCADA → L2 HMI display tamper (Manipulation of View).** WinCC SQL backend compromise (Stuxnet's path) or Citect Cicode runtime exploitation rewrites tag values shown to operators while real process data deviates. Operators see green when valves are open. [T0832 Manipulation of View](#) is the precondition for any L1 manipulation that must remain undetected long enough to cause physical effect — CRASHOVERRIDE and TRITON both depended on it.

Sources

- [CISA AA24-038A — Volt Typhoon US Critical Infrastructure](#)
- [CISA ICSA-21-056-03 — Rockwell Logix Controllers \(CVE-2021-22681\)](#)
- [CISA ICSA-21-110-06 — Eaton IPM](#)
- [CISA ICSA-21-294-02 — Honeywell Experion PKS upload / argument injection](#)

- [CISA ICSA-23-061-02 — Honeywell Experion PKS / LX / PlantCruise \(ICEFALL "Crystallized Insecurity"\)](#)
- [CISA ICSA-23-131-08 — SEL acSELeRator family](#)
- [CISA ICSA-23-193-01 — Rockwell FactoryTalk View ME](#)
- [CISA ICSA-23-194-02 — SEL Real-Time Automation Controller \(RTAC\)](#)
- [CISA ICSA-23-285-06 — Vertiv Avocent ACS 8000](#)
- [CISA ICSA-24-018-01 — Rockwell FactoryTalk Service Platform \(CVE-2024-21915\)](#)
- [CISA/NSA/FBI/ASD — Best Practices for Event Logging and Threat Detection \(2024\)](#)
- [Siemens ProductCERT advisory index](#)
- [Rockwell Automation Trust Center — Security Advisories](#)
- [Schneider Electric Cybersecurity Support Portal / PSIRT](#)
- [Honeywell PSIRT / Security Notifications](#)
- [SEL Security Advisories index](#)
- [Vertiv PSIRT / security advisory index](#)
- [Eaton PSIRT advisory index](#)
- [Claroty Team82 — Evil PLC Attack \(Rockwell + Schneider EcoStruxure Control Expert\)](#)
- [Claroty Team82 — Stealthy Rockwell PLC Hack \(CVE-2022-1161\)](#)
- [Claroty Team82 — ICEFALL Continues: Broken Trust, Broken Code \(Honeywell Experion\)](#)
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